📊 US Oncology Claims Analytics Pipeline – FRUZAQLA (fruquintinib)

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1. Objective & Business Context

Goal:
Analyze real-world **claims data** for **FRUZAQLA (Takeda)** in the US oncology market (metastatic colorectal cancer), benchmarking against competitors to evaluate utilization, adherence, and access patterns.

Key Questions:

1. Claim volume trends by state and payer type
2. Patient journey: initiation → continuation → discontinuation
3. Drug adoption in priority states (CA, NY, TX, FL)
4. Rolling utilization trends (R3M, R6M)
5. Provider-level prescribing behavior (new starts vs refills)
6. Payer mix impact (Commercial vs Medicare vs Medicaid)

Approach:

1. Raw claims → Standardization → Enrichment → Clinical + financial logic → Aggregated marts → Dashboards

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2. Architecture

Azure Blob Storage → Azure Data Factory → Azure SQL Database → Logic Apps → Power BI

- **Azure Blob Storage:** Raw claims ingestion (medical + pharmacy claims)
- **Azure Data Factory (ADF):** ETL pipelines (ingestion, transformation, orchestration)
- **Azure SQL Database:** Structured warehouse (layered model)
- **Logic Apps:** Pipeline monitoring, alerts, failure notifications
- **Power BI:** Business dashboards for commercial & analytics teams

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3. Pipeline Layers & Tables

Layer	Purpose	Tables (Claims-Focused)
LDG (Landing)	Raw ingestion	raw_claims_rx, raw_claims_px, eligibility_data, claim_headers, blob_qc_log
STG (Staging)	Cleaning & QC	clean_claims, qc_missing_patient, qc_invalid_ndc, qc_duplicate_claims, payer_mapping
FACT (Core Claims)	Atomic claims facts	claims_fact, patient_fact, provider_fact, drug_fact, payer_fact
INT (Intermediate)	Derived metrics	patient_journey, adherence_metrics, persistence_flags, rolling_utilization
MDM (Master Data)	Standard reference	drug_mdm, provider_mdm, payer_mdm, geography_mdm, competitor_mdm
BUS (Business Rules)	Domain logic	oncology_claims, line_of_therapy, new_vs_refill_flag, regimen_classification
RPT (Reporting)	Aggregated outputs	kpi_claims_monthly, patient_summary, provider_summary, payer_insights, state_summary

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4. Claims Data Model Considerations

- **Claim Types:**
  - Pharmacy (NDC-based drug claims)
  - Medical (procedure-based J-codes for oncology drugs)
- **Key Identifiers:**
  - Patient ID (de-identified)
  - Claim ID
  - Provider NPI
  - Payer ID
  - Service Date / Fill Date
- **Enrichment Logic:**
  - Map NDC → Drug
  - Identify FRUZAQLA vs competitors

- Link patient longitudinal history
- Classify line of therapy (LoT)

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5. Key KPIs (Claims-Specific)

KPI	Definition
Total Claims	COUNT(claim_id)
Unique Patients	COUNT(DISTINCT patient_id)
New Patient Starts (NPS)	Patients with first-ever claim for FRUZAQLA
Refill Rate	Refill claims / Total claims
Adherence (PDC)	Proportion of Days Covered
Persistence Rate	% patients continuing therapy after X months
Discontinuation Rate	Patients stopping therapy / Total patients
Avg Claims per Patient	Total claims / Unique patients
Payer Mix %	Claims split by payer type
State-wise Utilization	Claims per state
R3M Claims Trend	AVG(claims) over last 3 months
R6M Claims Trend	AVG(claims) over last 6 months
YoY Patient Growth	(Patients current year – last year) / last year
Provider Adoption Rate	Providers prescribing FRUZAQLA / Total oncology providers
Switch Rate	Patients switching from competitor drugs

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6. Business Logic Examples

- **New vs Refill Classification:**
  Based on first claim date per patient
- **Line of Therapy (LoT):**
  Derived using sequence of oncology treatments

- **Adherence Calculation:**
  Using claim gaps and days supply
- **Competitor Benchmarking:**
  Compare FRUZAQLA with peer drugs in same indication

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7. Dashboard Design (Power BI)

- **Line Charts:**
  - Claims trend (R3M, R6M) by drug
  - Patient growth over time
- **Bar Charts:**
  - Payer mix distribution
  - State-wise utilization
- **Maps:**
  - US heatmap showing patient density and claims
- **Tables:**
  - Top providers by new patient starts
  - High persistence providers
- **KPI Cards:**
  - Total Claims
  - Unique Patients
  - NPS
  - Adherence (PDC)
  - Persistence Rate
  - Payer Mix %
  - R3M / R6M Trends
  - Switch Rate

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8. Real-World Enhancements

- Integration with **specialty pharmacy data** for deeper adherence insights

- Inclusion of **diagnosis codes (ICD-10)** for precise cohort filtering
- Near real-time ingestion for faster commercial decision-making
- Automated alerts via Logic Apps for sudden drop in claims or adherence

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9. Pipeline Characteristics

- **Execution Frequency:** Weekly
- **Latency:** ~1–2 hours
- **Scalability:** Handles millions of claims records
- **Data Quality:** Multi-level QC checks (nulls, duplicates, invalid codes)
- **Governance:** HIPAA-compliant, de-identified patient data

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